

Signal & Data Analysis in Neuroscience
2020
Lesson 7: Neural Discrimination

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Discrimination

- Discrimination is the discernment of qualities and recognition of the differences between things (wikipedia)

■ הבחנה, הבדלה; סיווג (מורפיקס)

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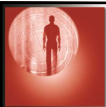
The neuronal transformation

$s(t) \rightarrow \text{Sensors} \rightarrow r(t) \rightarrow \text{Spike Generator} \rightarrow \rho(t)$
 Event related potential
 Neural encoding

$s(t) \leftarrow \text{Sensors} \leftarrow r(t) \leftarrow \text{Spike Generator} \leftarrow \rho(t)$
 Neural discrimination
 Neural decoding

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Outline

- Discrimination
- ROC curves
- Likelihood ratio

Based upon: Theoretical neuroscience, *Dayan P & Abbott LF*, Ch. 3
(However, using the complementary definition of β)

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Dot movement discrimination





- Stimulus is moving dot pattern with variable percentage of coherently moving dots.
- Monkey behavioral forced choice task to report direction of motion (up or down).
- Can the movement be discriminated?
 - The monkey's behavior.
 - The information conveyed by a single neuron (area MT).

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Based on: Britten, K. H., Shadlen, M. N., Newsome, W. T., & Movshon, J. A. (1992)

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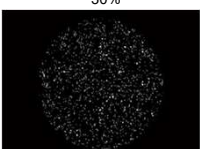
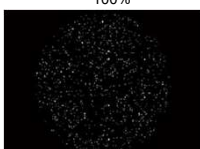



Movement discrimination demo

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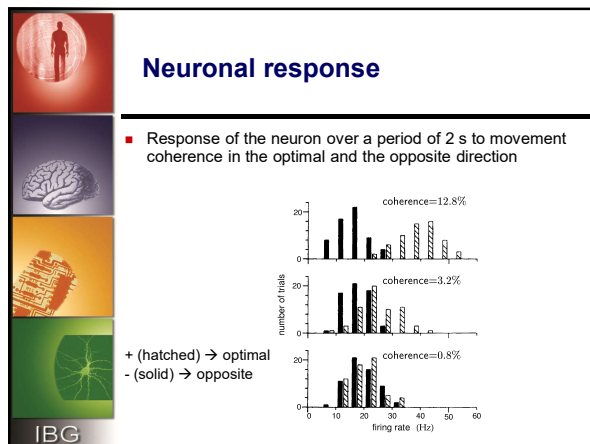



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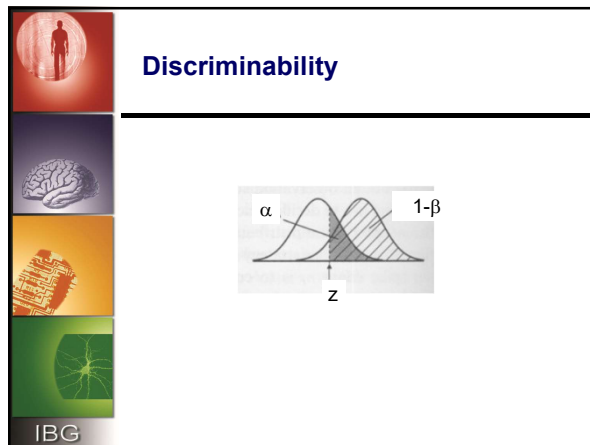



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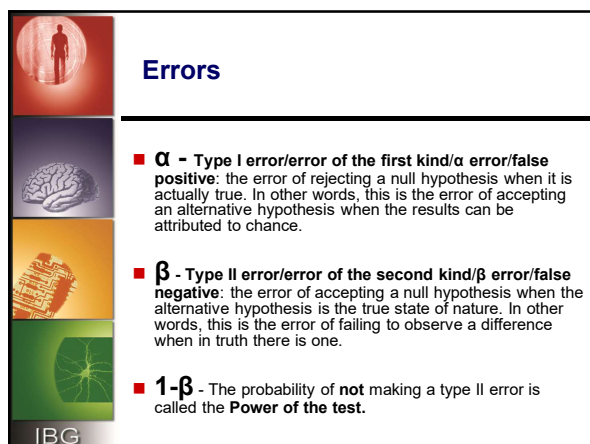
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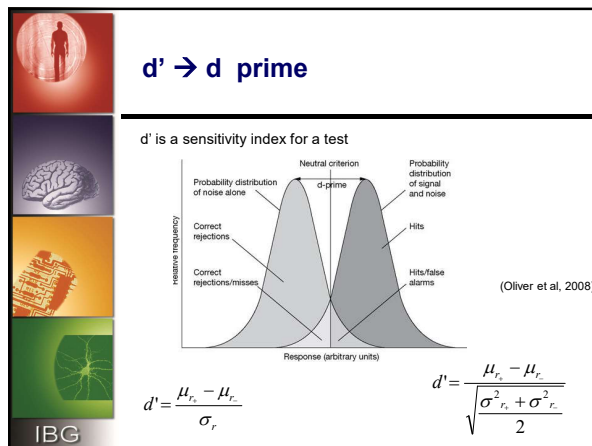
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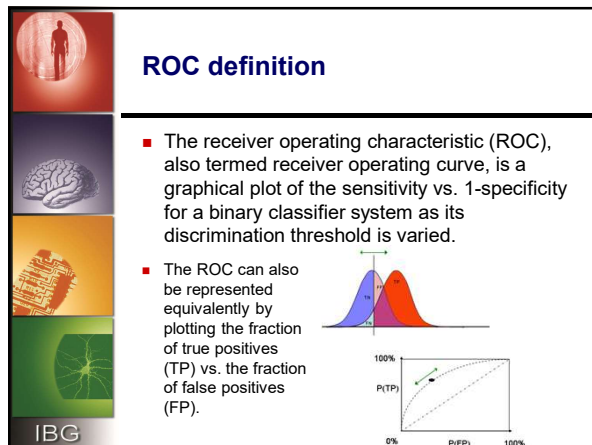
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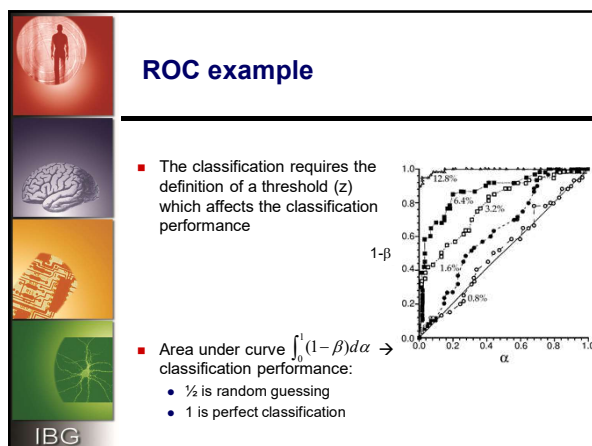
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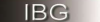





ROC tutorial

<http://arogozhnikov.github.io/2015/10/05/roc-curve.html>
<http://www.navan.name/roc/>

Demo



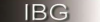
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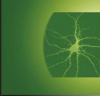

Two alternative forced choice I

- Different thresholds (z) may be chosen
- This ambiguity is removed by performing a *two alternative forced choice* – presentation of the two options and choosing which one represented which stimulus.
- The choice is made according to the rates r_1 , r_2 where r_1 serves as the threshold for r_2



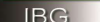
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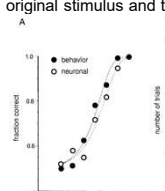
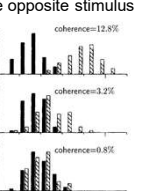


Two alternative forced choice

- Neuronal response is as if based on two alternative forced choice viewed by ideal observer
- The observer is given the hypothetical response to the neuron and anti-neuron
- In reality the observer views the response to the original stimulus and the opposite stimulus



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The likelihood (גיראות) function I

- A likelihood function is a conditional probability function considered a function of its **second** argument with its **first** argument held fixed. For a specific b the likelihood is $P(A|B=b)$.
- Probability allows us to predict unknown outcomes based on known parameters, while likelihood allows us to estimate unknown parameters based on known outcomes.
- It is also any other proportional function $L(b|A) = \alpha P(A|B=b)$ for every $\alpha > 0$
- Thus, the numerical value of L is not important but only the **ratio** of likelihoods: $\frac{L(b_1|A)}{L(b_2|A)}$

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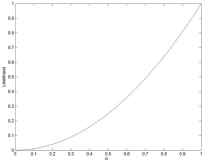
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The likelihood function II

- It is important to remember that the likelihood of a given b is **not** the same as the probability of a given b .
- The likelihood function is **not** a probability density function e.g. the integral of a likelihood function is not in general 1.

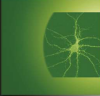


Likelihood of the coin probability given two observations "HH" (Wikipedia)

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Likelihood ratio & Neyman-Pearson Lemma

- Classification based on r (in the previous case) gave good results. Is it possible to do **better**?
- Likelihood ratio ($s=\pm$): $l(r) = \frac{p(r|+)}{p(r|-)}$
- Neyman-Pearson lemma: the likelihood ratio provides the optimal discriminator
- Any other monotonic function of the likelihood ratio will provide as good a test as the ratio itself
 - In the previous example $l(r)$ is monotonically increasing with r
 - In many cases the log of $l(r)$ is used.

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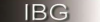

The likelihood ratio test

$$\alpha(z) = \int_{-\infty}^{\infty} p(r|-)dr \quad \rightarrow \quad \frac{d\alpha}{dz} = -p(z|-)$$

$$\beta(z) = \int_{-\infty}^{\infty} p(r|+)dr \quad \rightarrow \quad \frac{d\beta}{dz} = -p(z|+)$$

$$\frac{d\beta}{d\alpha} = \frac{d\beta}{dz} \frac{dz}{d\alpha} = \frac{p(z|+)}{p(z|-)}$$

- The slope of the ROC curve is equal to the likelihood ratio.



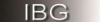
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Population discrimination

- **Population encoding** – individual neurons in the population have different but overlapping selectivities so that many neurons respond to a single stimulus.
- Useful for:
 - Redundancy
 - Reduction of variability
 - Encoding multiple aspect of a stimulus
- Discrimination can be performed exactly the same way by using **r (vector)** instead of r (scalar) for each response.



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